Statistical Thinking for Data Science

Week 10: Testing claims, and the trouble with p-values

A decision rule, and everything it is not

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Week 10: Testing claims, and the trouble with p-values

Learning Objectives

By the end of this week you should be able to:

- set up a null and an alternative hypothesis for a practical question;
- carry out a one sample t test and a two proportion test, and read the output;
- define a p-value exactly, and list four things it is not;
- distinguish statistical significance from practical importance, and report an effect size with an interval;
- explain type I and type II errors, power, and how sample size is chosen;
- recognise p-hacking and the multiple comparisons problem, and say what pre-registration fixes.

10.1 The logic of a test

A test is a decision rule for a yes or no question about a population, made from a sample. The structure never changes:

1. State a **null hypothesis** H_0, the boring explanation: no difference, no effect, the target is being met.
2. State an **alternative** H_1, what you are prepared to conclude instead.
3. Choose a **test statistic** that measures how far the data sit from H_0, in standard error units.
4. Compute the **p-value**: the probability, *if* H_0 *were true*, of a test statistic at least as extreme as the one you got.
5. Decide, using a threshold fixed in advance, and report the effect with its interval either way.

The null hypothesis is a device, not a belief. It is the statement under which you can compute what the data should look like, which is why it is always the one with the equals sign.

10.2 Two standard tests

Example 1: is the grade above the cut-off?

Twenty five cores give $\bar{x} = 2.41$ g/t and $s = 0.62$. The mine plans on a mean grade of 2.2 g/t. Is the block better than plan?

Solution

$H_0 : \mu = 2.2$ against $H_1 : \mu \neq 2.2$. SE $= 0.62/\sqrt{25} = 0.124$, so

$$t = \frac{2.41 - 2.2}{0.124} = 1.69 \quad \text{on 24 degrees of freedom,}$$

and the two-sided p-value is 0.103.

At the conventional 5% threshold this is not significant. The 95% interval from Week 9 was (2.15, 2.67), which includes 2.2, and says the same thing with more information: the block may be as poor as plan or as rich as 2.67, and 25 cores cannot separate those.

What not to write

"There is no difference between the block and plan." Failing to reject is not evidence of equality. Write: "the data are consistent with the planned grade; a true mean anywhere from 2.15 to 2.67 g/t is compatible with what we drilled."

Example 2: two suppliers of concrete cubes

Supplier A: 12 failures in 180. Supplier B: 24 in 220.

Solution

$\hat{p}_A = 0.0667$, $\hat{p}_B = 0.1091$, pooled $\hat{p} = 0.09$.

$$z = \frac{0.1091 - 0.0667}{\sqrt{0.09(0.91)\left(\frac{1}{180} + \frac{1}{220}\right)}} = 1.48, \qquad p = 0.140.$$

Not significant at 5%. The difference in failure rates is 4.2 percentage points with a 95% interval of $(-1.3,\ +9.7)$ points.

The honest summary

"Supplier B failed more often, 10.9% against 6.7%, but with 400 cubes the difference is not distinguishable from chance: the data are consistent with anything from B being slightly better to B being ten points worse." That sentence supports a decision to keep testing, which is the correct decision, rather than either cancelling the contract or declaring the suppliers equal.

10.3 What a p-value is, and four things it is not

The p-value is $P(\text{data at least this extreme} \mid H_0 \text{ true})$. It is a conditional probability of *data*, given a hypothesis.

It is **not**:

- the probability that H_0 is true. That is $P(H_0 \mid \text{data})$, the other direction, and Week 8 showed what happens when the two are swapped. Getting it requires a prior.
- the probability that your result was due to chance.
- a measure of the size or importance of an effect. With 100 000 observations, a difference of no practical consequence has $p < 0.001$.
- the probability of replication. A study with $p = 0.04$ is nowhere near 96% likely to repeat.

How often is a significant result true?

Suppose one hypothesis in ten that you test is really true, your tests have 80% power, and you work at the 5% level. Of 1 000 hypotheses, 100 are true and 80 are detected; 900 are false and 45 come out significant anyway. So 125 significant results, of which 80 are real: 64%. With weaker power, say 30%, it falls to 40%, and most of your significant findings are wrong. This is the Bayes calculation from Week 8 with hypotheses in place of patients, and it explains the replication crisis more simply than any accusation of fraud.

10.4 Errors, power and sample size

	H_0 **true**	H_0 **false**
Reject H_0	type I error, probability α	correct, probability $1-\beta$ (power)
Do not reject	correct	type II error, probability β

α is chosen, conventionally 0.05, and 0.05 is a convention and nothing more. Power depends on the sample size, the variability, and how big the effect really is. For comparing two means with a standardised effect size d,

$$n \text{ per group} \approx \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2}{d^2}.$$

For $d = 0.5$ at 80% power and 5%: $n = 2(1.96 + 0.84)^2/0.25 = 62.8$, so 63 per group. For a small effect, $d = 0.2$, it is 393 per group. Deciding this *before* collecting data is the difference between a study that can answer its question and one that cannot.

An underpowered study is not a cheap study

If the design has 30% power, a null result tells you almost nothing, and a significant result is more likely to be a false positive than a real finding. The money was spent to produce an uninterpretable answer. Power is not a statistical nicety: it is whether the study was worth doing.

10.5 Multiplicity and p-hacking

Test twenty independent true nulls at 5% and the chance of at least one significant result is $1 - 0.95^{20} = 0.64$. Significance is nearly guaranteed if you test enough things, which is why the following are all forms of cheating, whether or not anyone intends it:

- trying several outcome variables and reporting the one that worked;
- splitting the sample by programme, gender or shift until a subgroup is significant;
- collecting data until $p < 0.05$ and then stopping;
- dropping "outliers" only when they spoil the result;
- writing the hypothesis after seeing the answer, which is undetectable in the final paper.

Remedies, in increasing order of strength: report every test you ran, not the survivors; adjust the threshold, for example Bonferroni's α/k, so twenty tests use 0.0025 each; split the data and confirm on the held-out half; and best of all, write the analysis plan down before the data arrive. For your group project, the protocol you submit *is* your pre-registration, and I will compare it with your report.

10.6 Reporting a test properly

A complete result is: the estimate, its interval, the test statistic, the p-value, the sample size, and the pre-specified threshold. For example:

> Failure rates were 6.7% (A) and 10.9% (B), a difference of 4.2 percentage points (95% CI -1.3 to $+9.7$; $z = 1.48$, $p = 0.14$; $n = 180$ and 220). This was the single pre-specified comparison. The data do not distinguish the suppliers at the 5% level, and a difference of up to about ten points remains compatible with them.

Report the interval first and the p-value last. The interval tells the reader how big the effect might be, which is the question somebody actually has to act on.

10.7 Doing it on a computer

R: `t.test`, `prop.test`, `chisq.test`, `power.t.test`. Python: `scipy.stats.ttest_1samp` and `ttest_ind`, `statsmodels.stats.proportion.proportions_ztest`, and `statsmodels.stats.power` for sample size. Any of these will print a p-value for nonsense input, so the thinking is yours: the software checks nothing about the design.

10.8 Glossary

Null hypothesis	the statement under which the sampling distribution can be computed
Test statistic	the distance between data and H_0, in standard error units
p-value	$P(\text{data this extreme} \mid H_0)$
Significance level α	the type I error rate chosen in advance
Power	the probability of detecting a real effect of a stated size
Effect size	the size of the difference, in units or standardised
Bonferroni correction	dividing α by the number of tests
Pre-registration	recording the analysis plan before seeing the data

10.9 Summary

Week 10 in six lines

1. A test is a decision rule, and the null is a device, not a belief.
2. A p-value is the probability of the data given H_0, nothing else.
3. Not significant does not mean no effect; it often means not enough data.
4. Significant does not mean important; report the effect size and interval.
5. Power decides whether the study was worth running, and it is set in advance.
6. Twenty tests at 5% give a 64% chance of a false positive, so say in advance what you will test.

Exercises

1. Twenty five cores give $\bar{x} = 2.41$, $s = 0.62$, against a planned grade of 2.2 g/t. Carry out the test, report it properly, and write the one sentence you would put in the mine's weekly report.

2. Suppliers A and B fail 12 of 180 and 24 of 220 cubes. Test the difference, give a confidence interval for it, and state the decision you would recommend and why.

3. A colleague reports $p = 0.049$ and writes "there is a 95% probability the treatment works". Write two sentences correcting him, without using the word "wrong".

4. A study of 40 000 customers finds that a new app layout increases session time by 1.2 seconds, $p < 0.001$. Should the company rebuild the app? What question does the p-value fail to answer?

5. You plan to compare two groups and expect a standardised effect of about 0.5. How many per group do you need for 80% power at the 5% level? What happens to that number if the real effect is 0.2?

6. A team tests 20 outcome variables and reports the three with $p < 0.05$. Compute the probability of at least one false positive among 20 true nulls, and say what you would ask the team for.

7. Explain why a null result from an underpowered study is nearly uninterpretable, using a numerical example of your own.

8. Write the analysis plan for your group project in one paragraph: the comparison, the outcome, the test, the threshold, and the sample size you will collect. This is your pre-registration.

Answers to the odd-numbered exercises

1. $t = (2.41 - 2.2)/(0.62/5) = 1.69$ on 24 df, $p = 0.103$, 95% CI $(2.15, 2.67)$. Report: "mean grade 2.41 g/t (95% CI 2.15 to 2.67, $n = 25$); the block is not distinguishable from the planned 2.2 g/t at the 5% level."

3. "The 0.049 is the chance of seeing data like ours if the treatment did nothing, not the chance that the treatment works. To get the second number you would also need to say how plausible the treatment was before the study, and with a modest prior and typical power a result at $p = 0.049$ leaves a substantial chance of being a false positive."

5. $n = 2(1.96 + 0.84)^2/0.5^2 = 62.8$, so 63 per group. At $d = 0.2$ it is $2(2.80)^2/0.04 = 393$ per group, six times as many, because the required sample grows with the square of the inverse effect size.

7. With 30% power, a true effect is missed 70% of the time, so a null result is the expected outcome whether or not the effect exists, and it discriminates almost nothing. Example: 20 per group looking for $d = 0.3$ has power near 0.13, so failing to find significance is close to guaranteed in advance.

Before next week

Bring the analysis plan from exercise 8. Next week we fit models: the straight line, what adjustment means when you add a second predictor, and the difference between predicting and explaining.